

Lecture 11:

Gaussian Processes

September 30, 2026

we now know how to sample high dimension distributions:
rejection sampling, importance sampling, MCMC, Hamiltonian MC

let's apply it to astro! choose your adventure:

ast384c/inclass/lec9_hierarchical/review_emission.ipynb

ast384c/inclass/lec9_hierarchical/review_sed.ipynb

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what if you don't have a model?

Machine Learning: Gaussian Processes

goal of machine learning: learn a function from examples
without a mechanistic model of what produced the data

using a flexible function class with many parameters

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gaussian processes: ML you can still do **by hand**

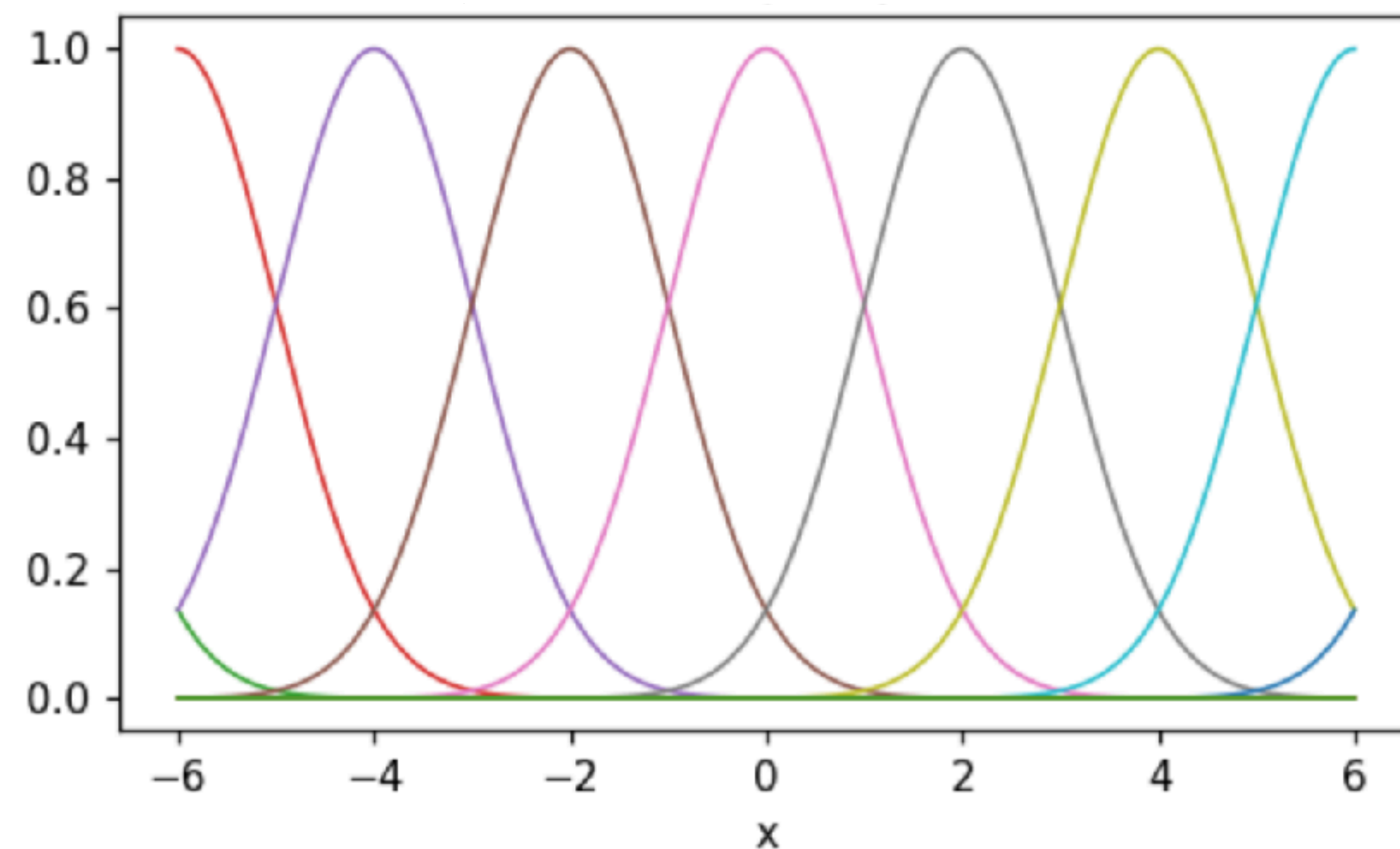
gaussian processes: functions as random variables

$$f(x) = \sum_{m=1}^M w_m \phi_m(x) = \boldsymbol{\phi}(x)^\top \mathbf{w}$$

$$\phi_m(x) = \exp \left[-\frac{(x - c_m)^2}{2\ell^2} \right]$$

gaussian processes: functions as random variables

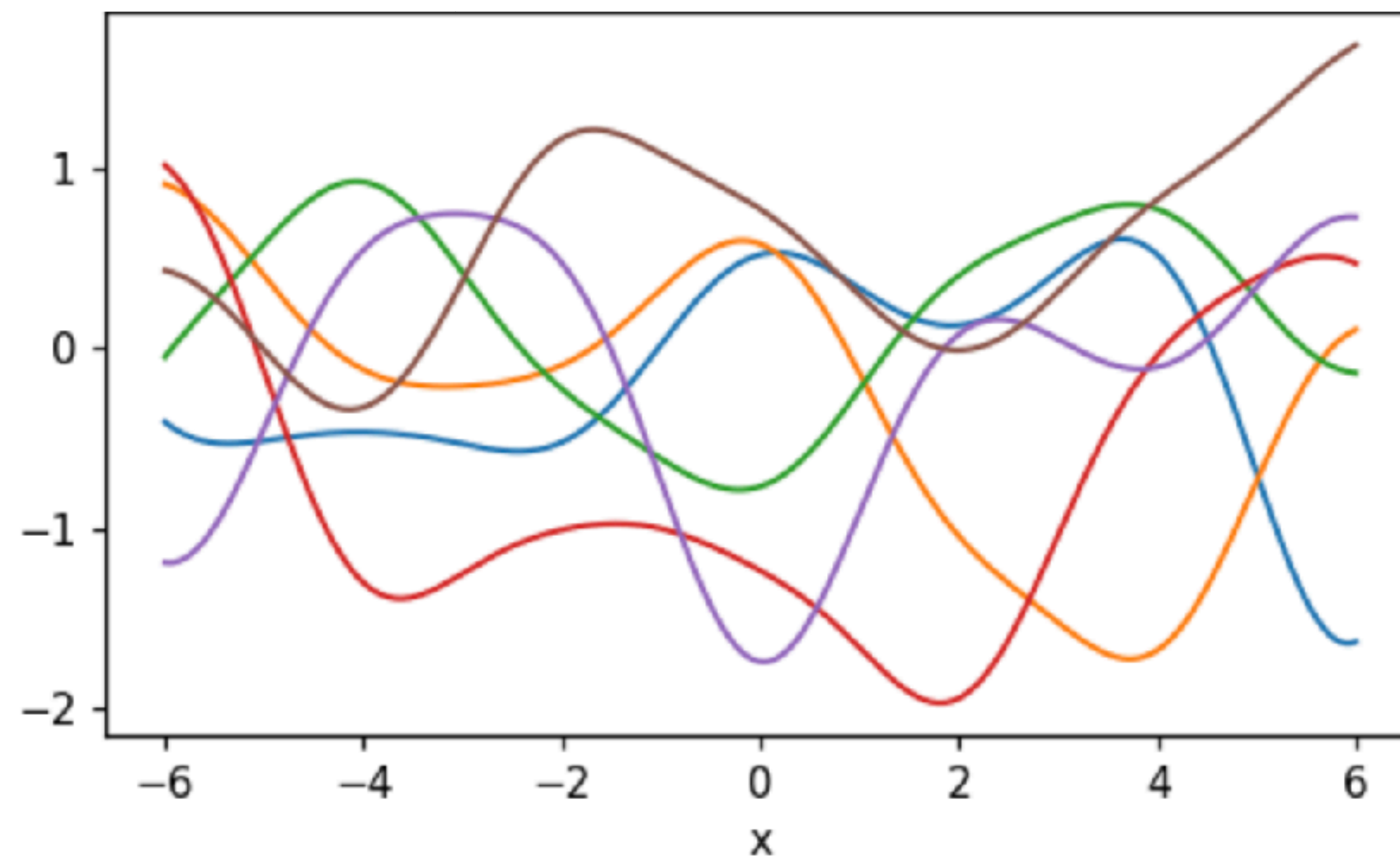
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$$w_m = 1$$

gaussian processes: functions as random variables

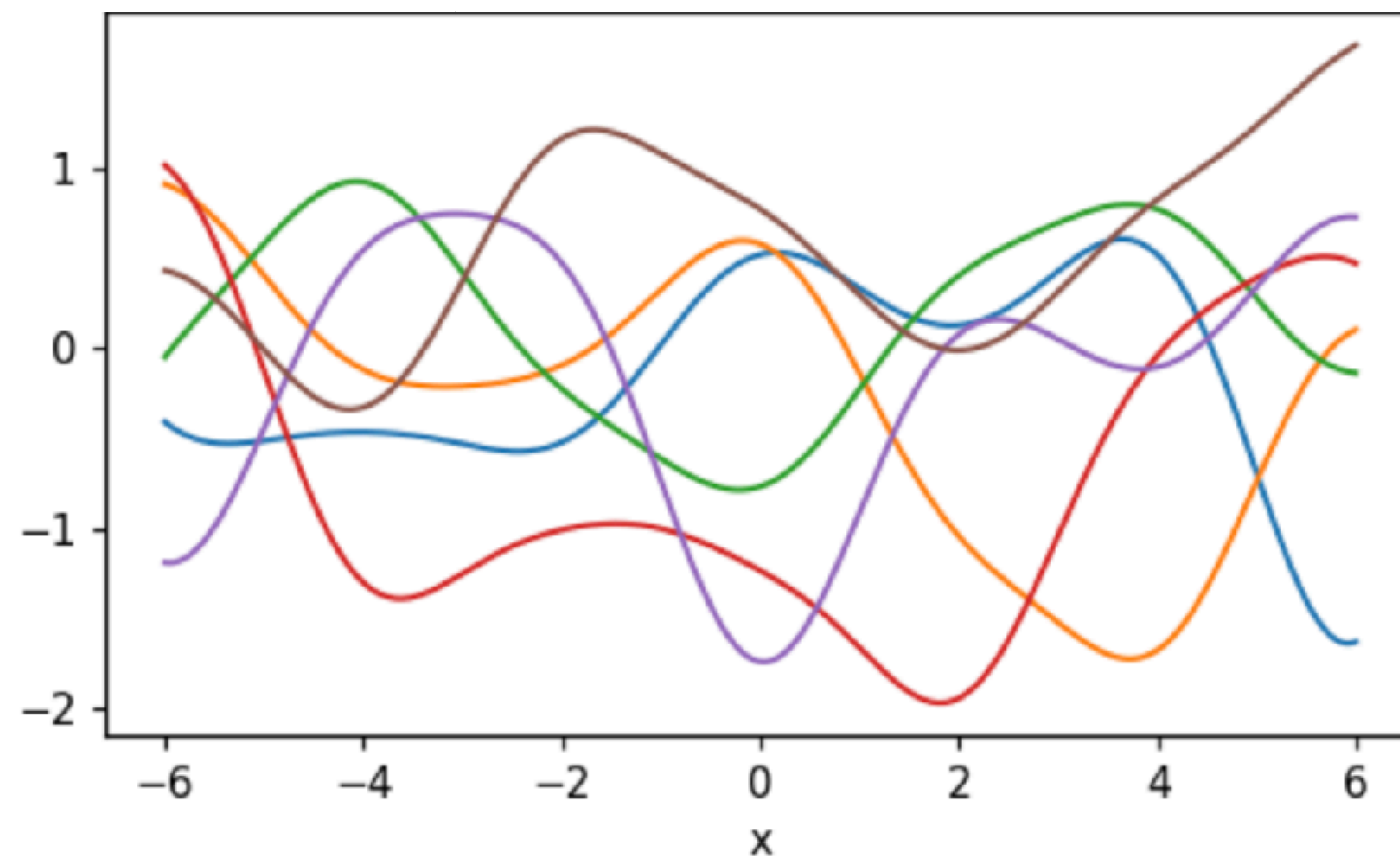
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a prior on $w =$
a prior on functions

we don't care about w so let's marginalize it out

$$\mathbb{E}[f(x)] = \int f(x; w) p(w) \, dw = \boldsymbol{\phi}(x)^\top \mathbb{E}[\mathbf{w}] = 0$$

$$\text{Cov}[f(x), f(x')] = \boldsymbol{\phi}(x)^\top \mathbb{E}[\mathbf{w}\mathbf{w}^\top] \boldsymbol{\phi}(x') = \sigma_w^2 \boldsymbol{\phi}(x)^\top \boldsymbol{\phi}(x') \equiv k(x, x')$$

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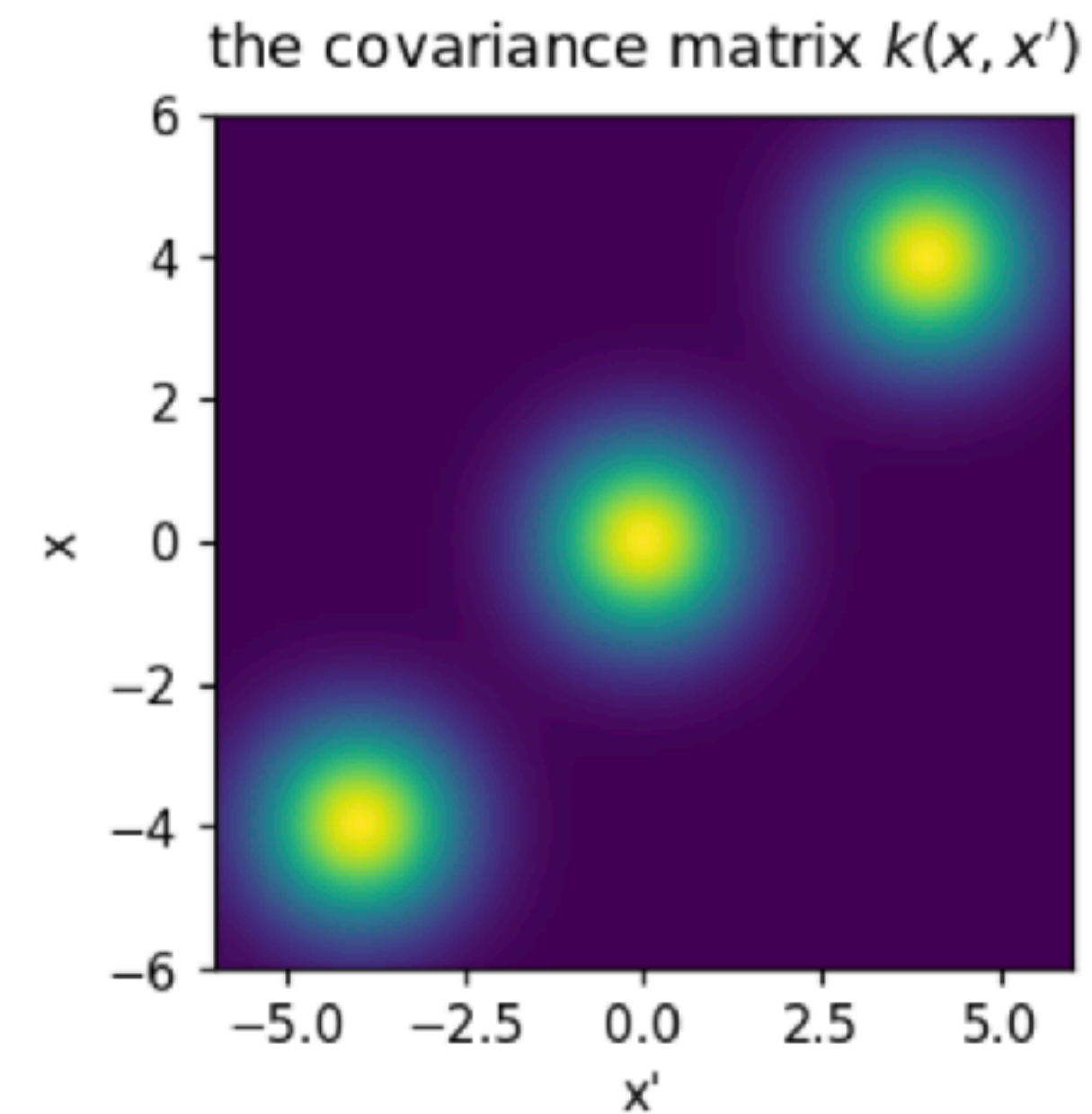
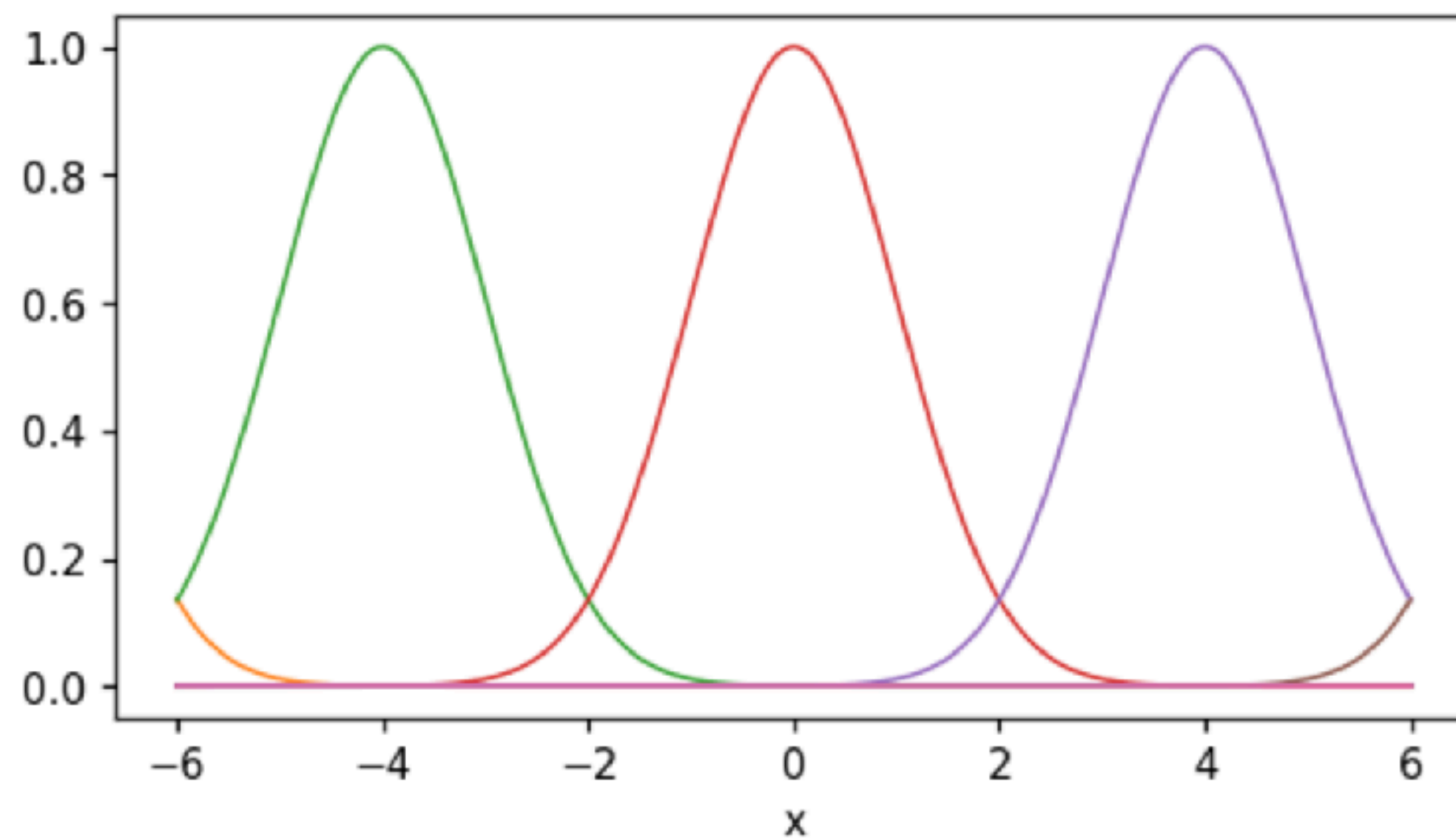
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as we decrease $\Delta c = c_m - c_{m-1} \rightarrow 0$

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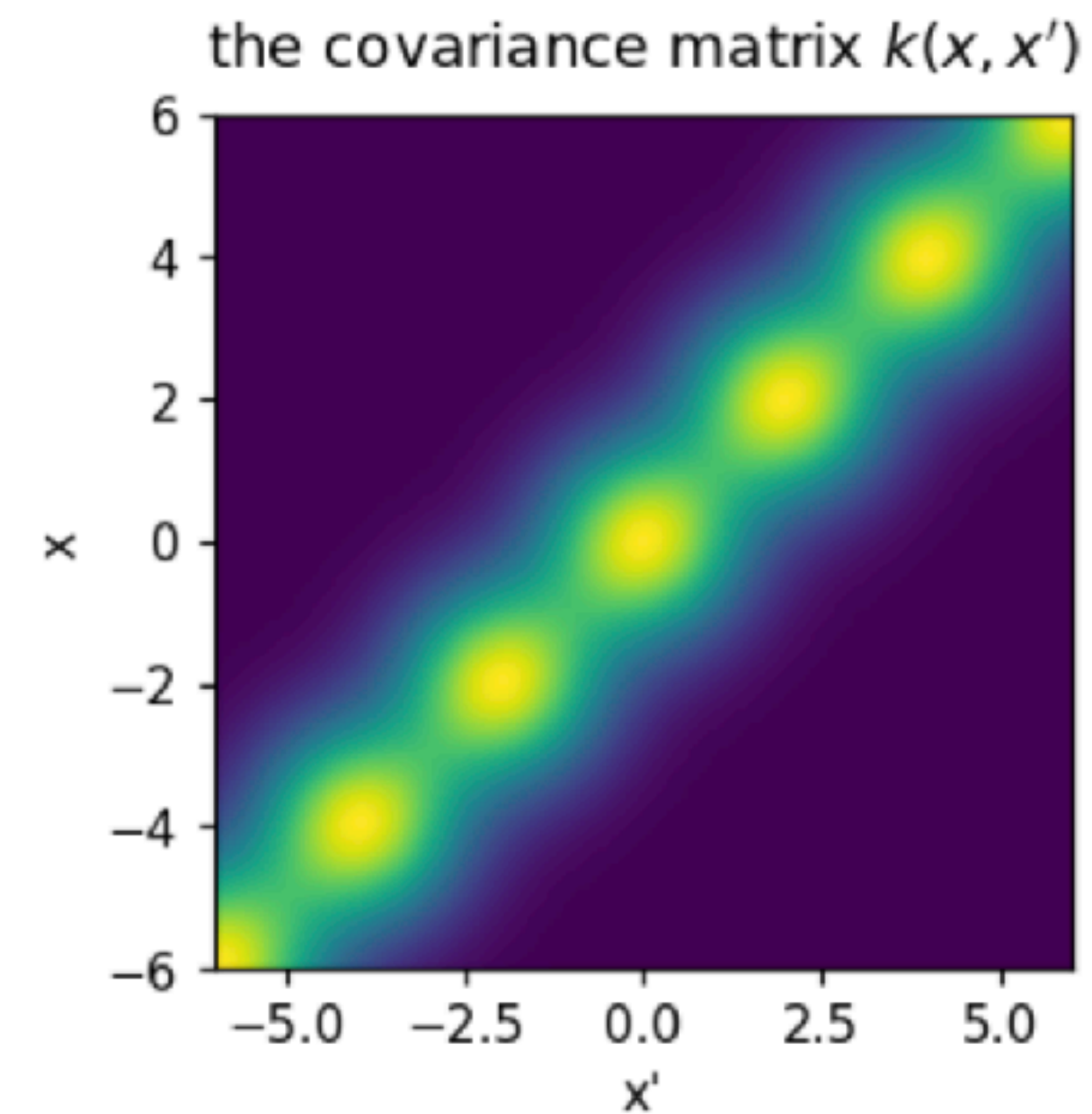
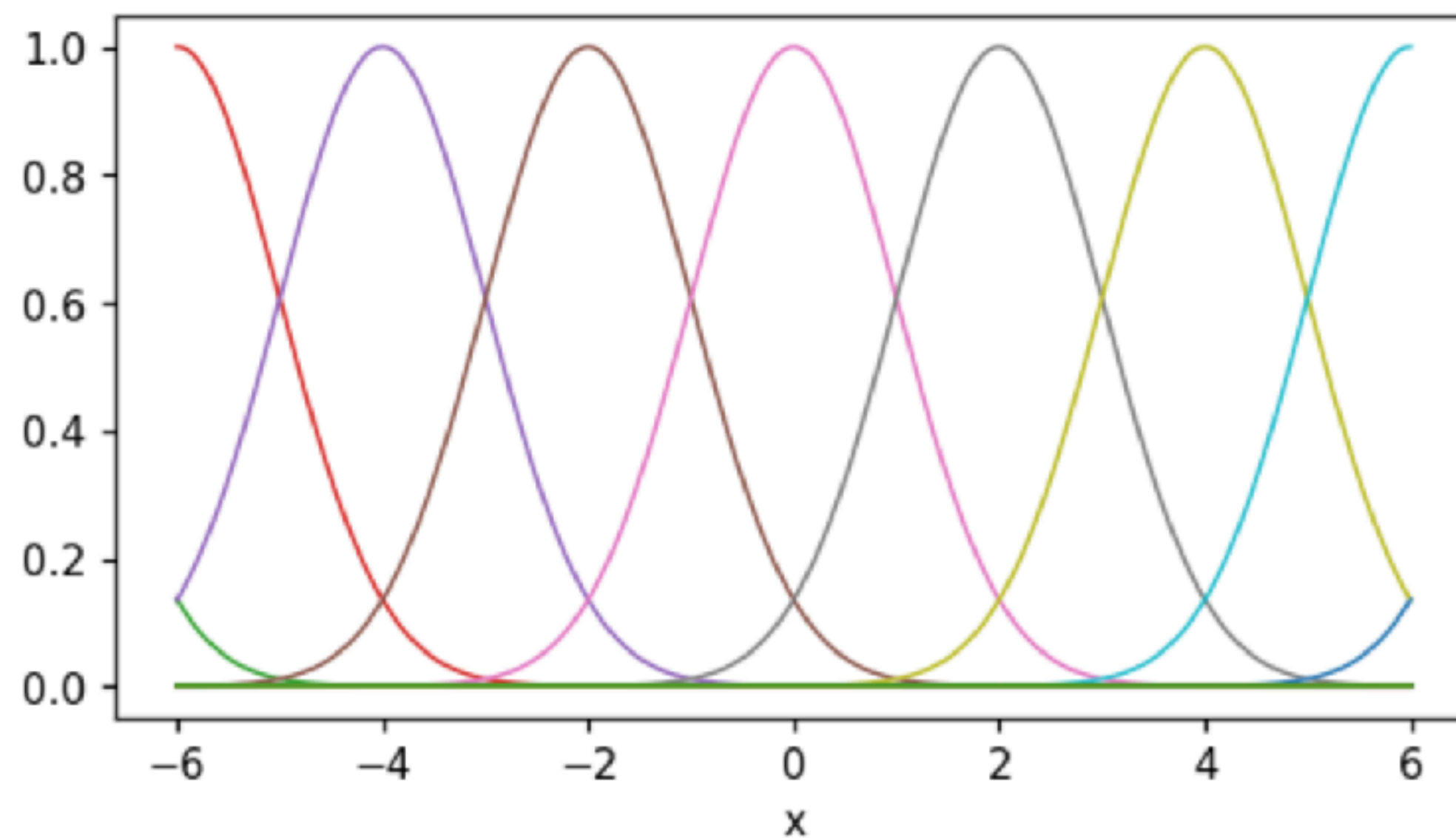
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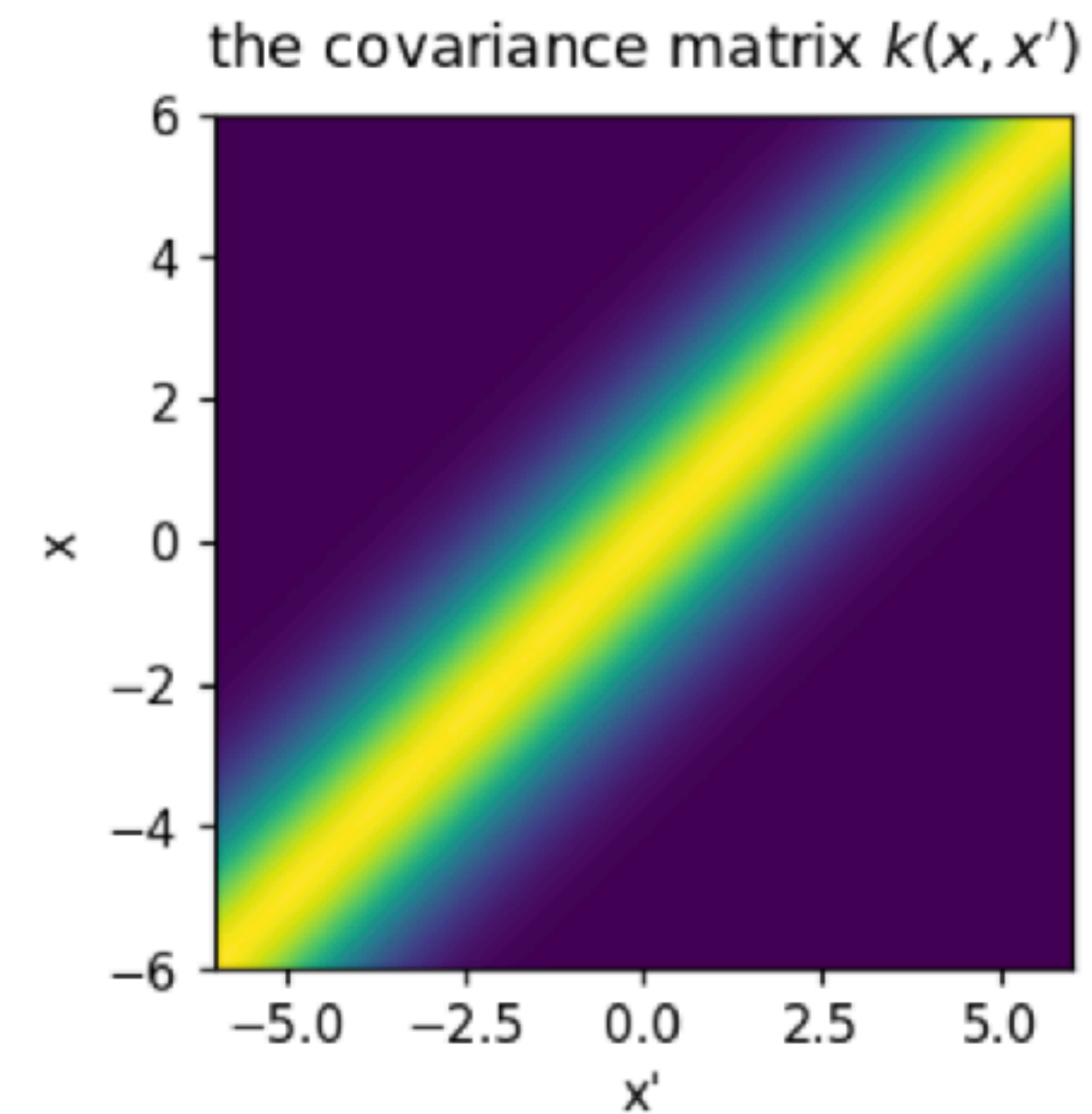
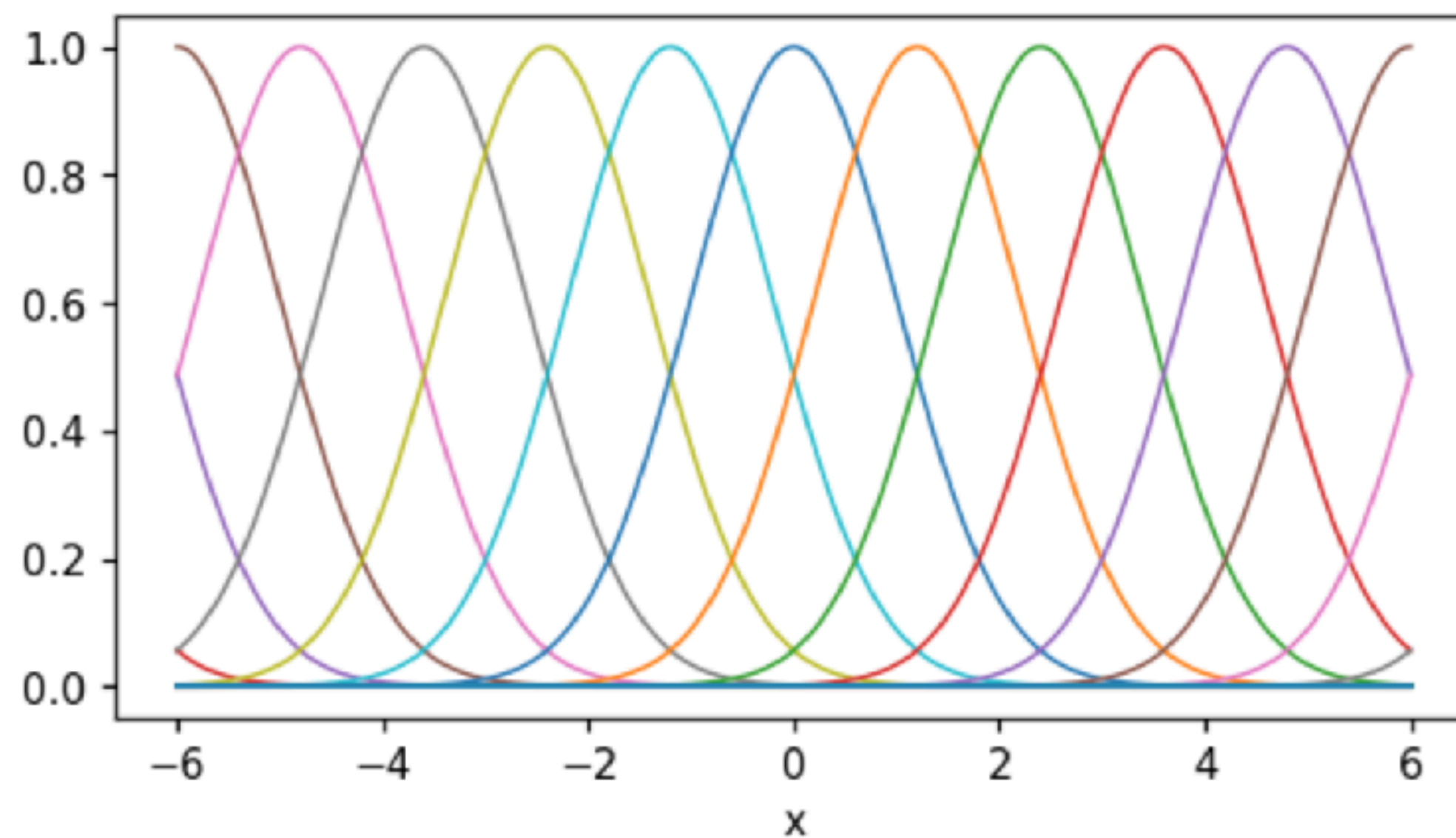
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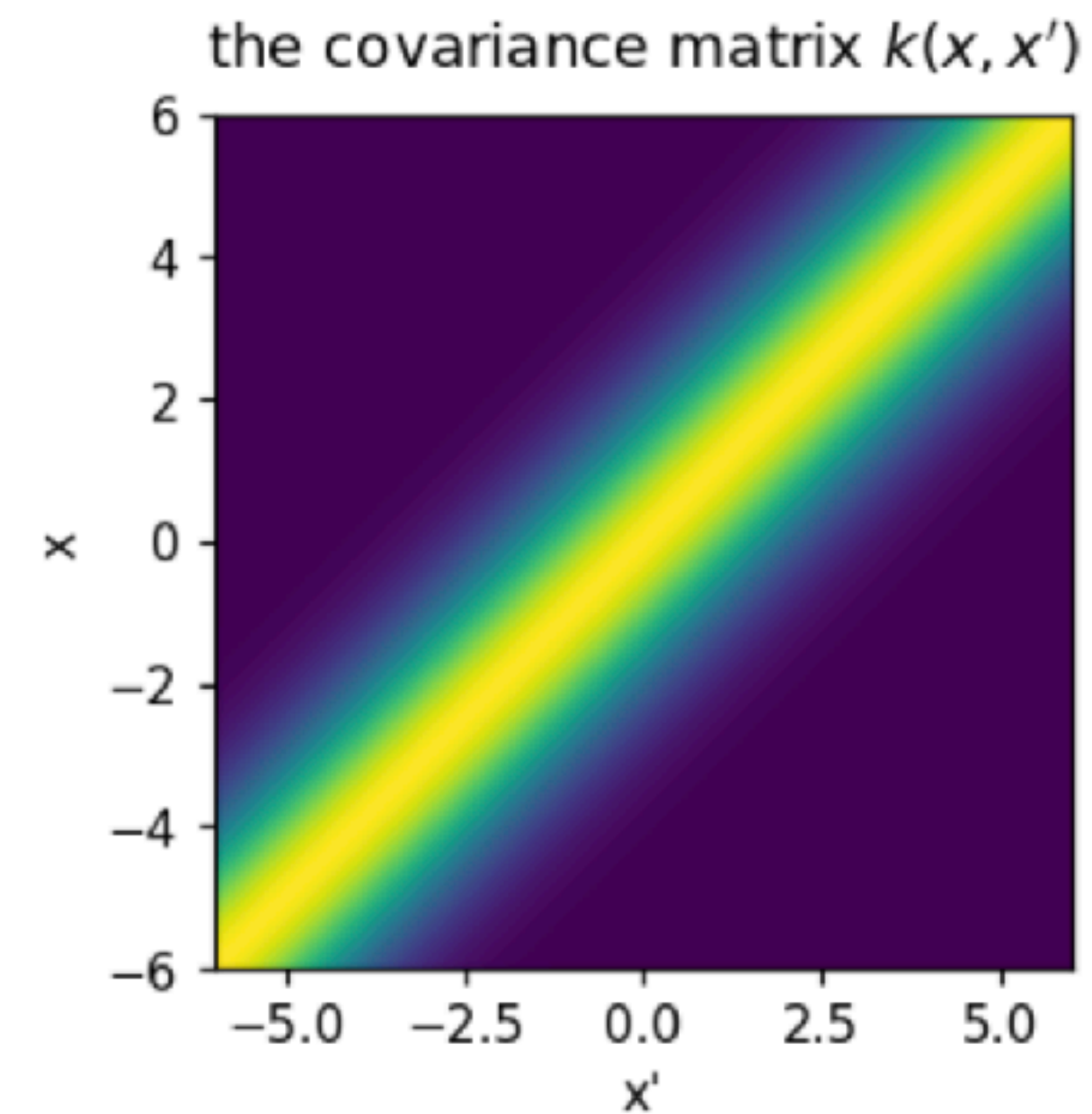
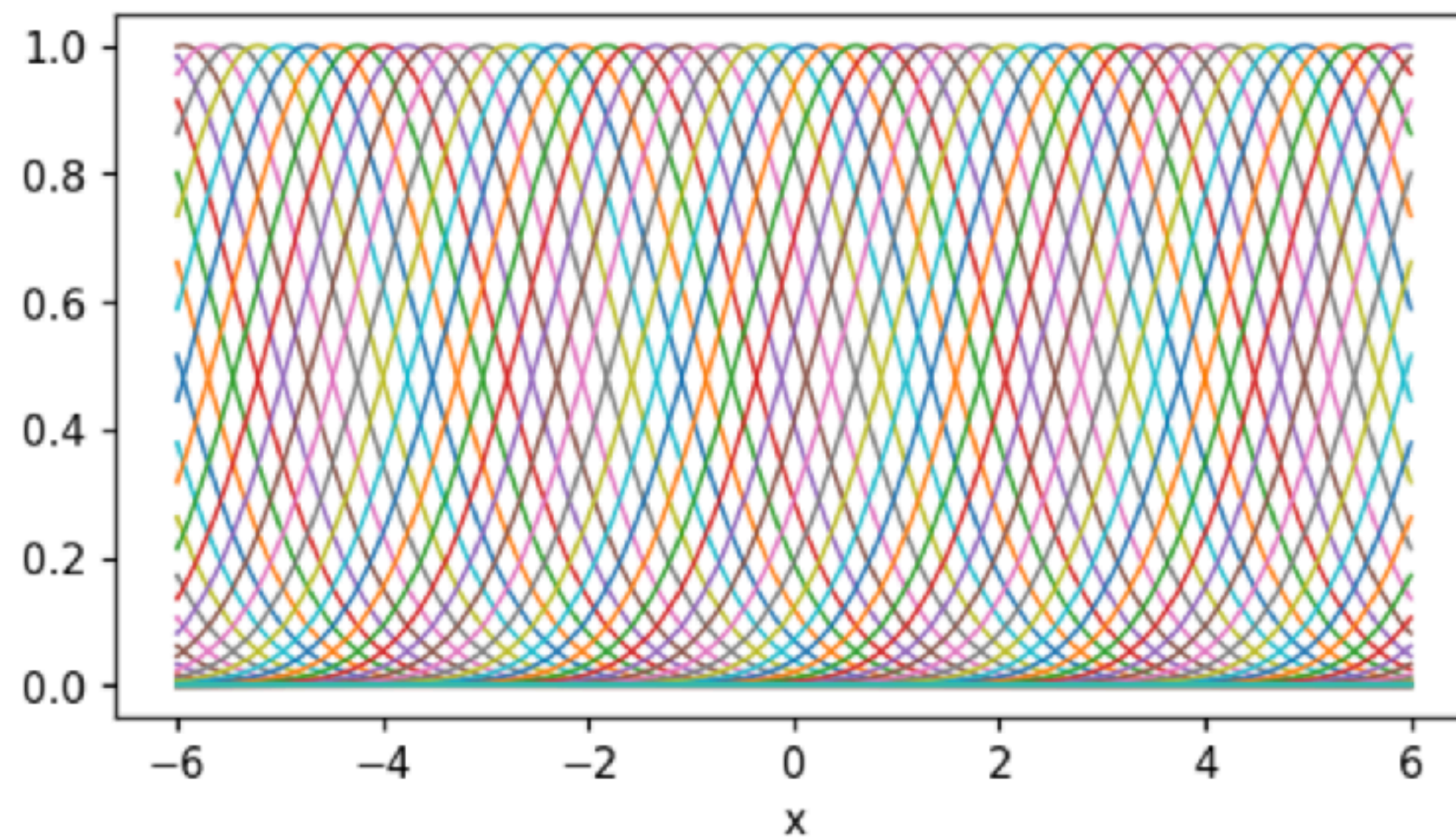
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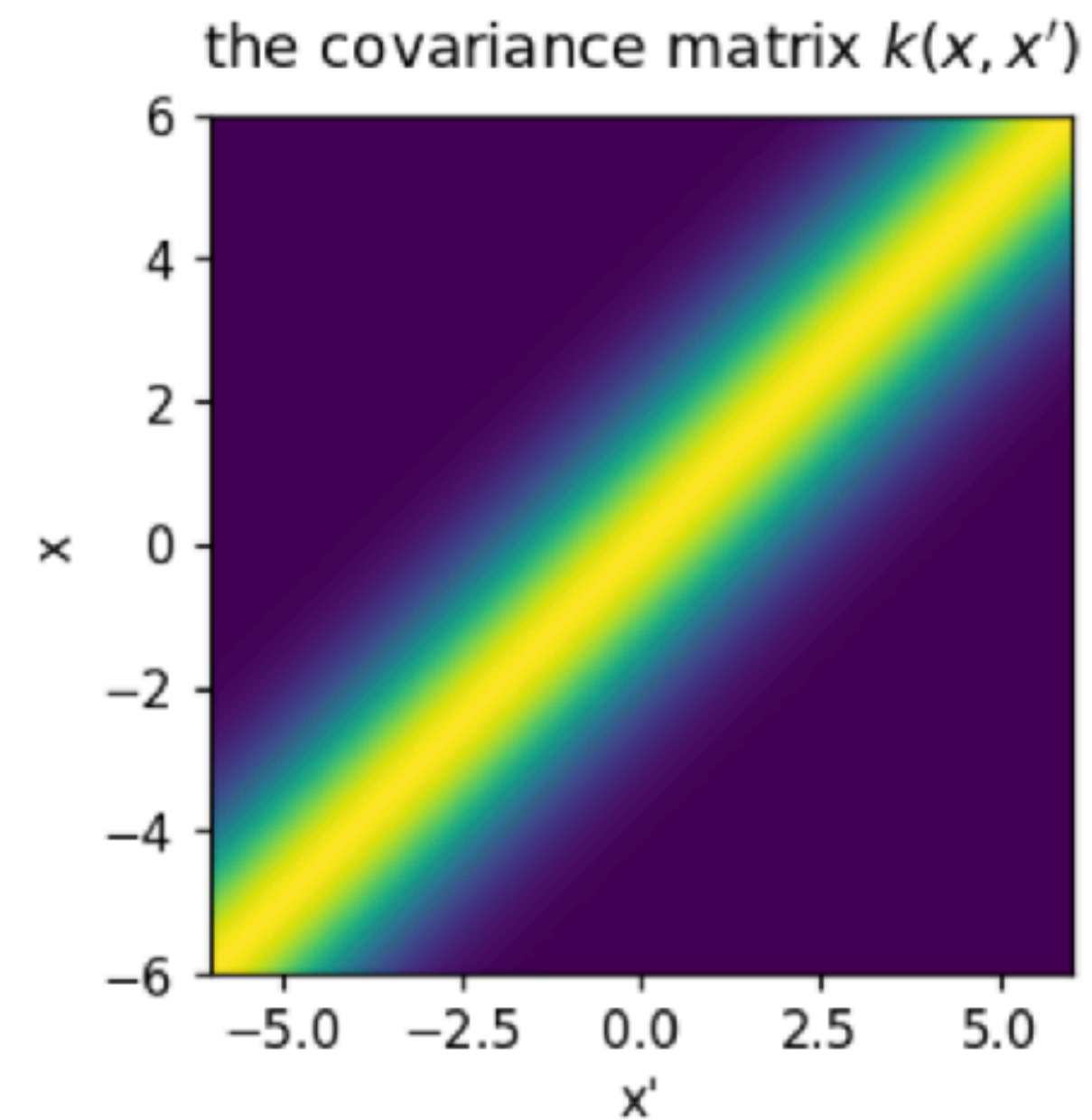
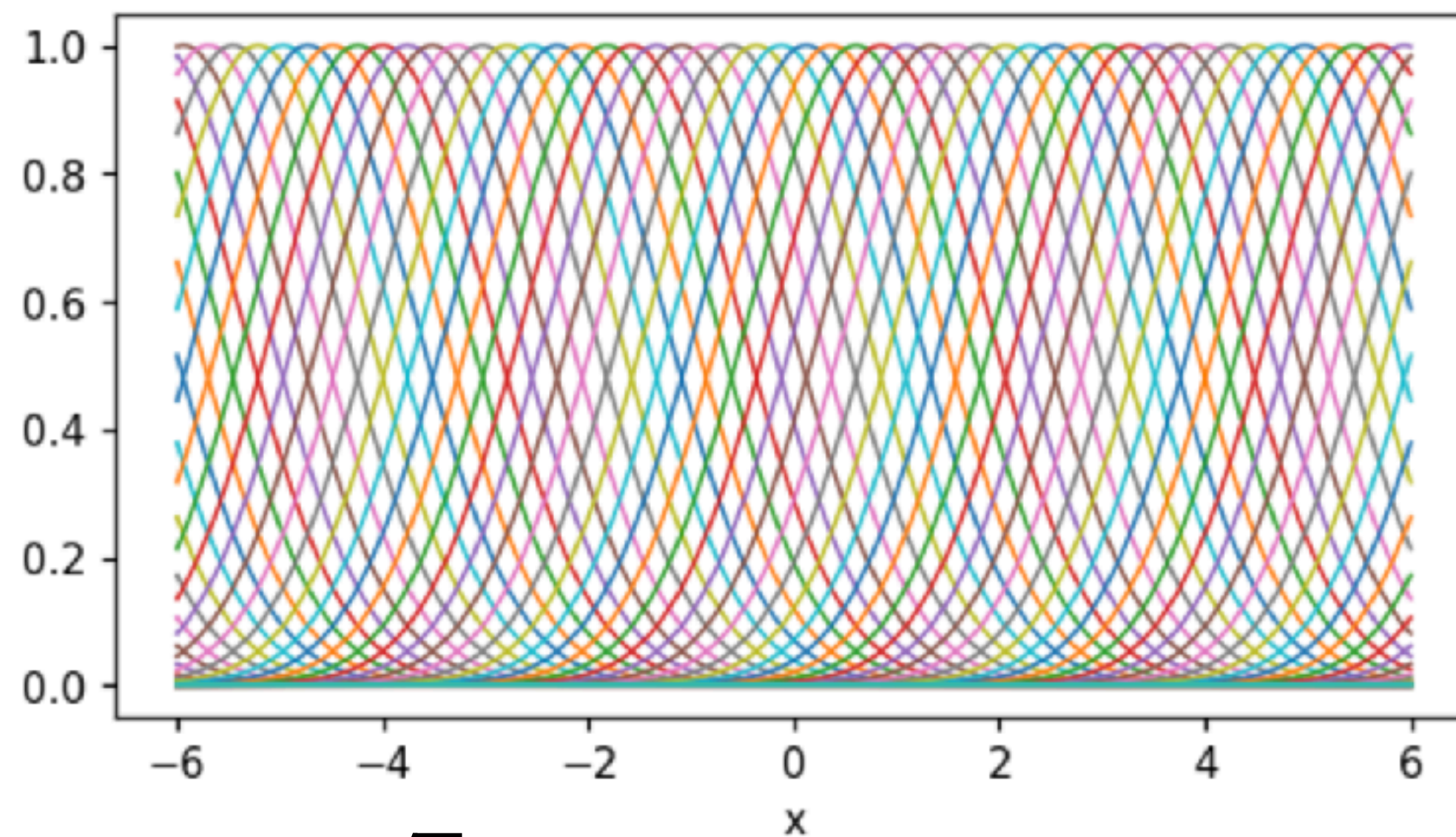
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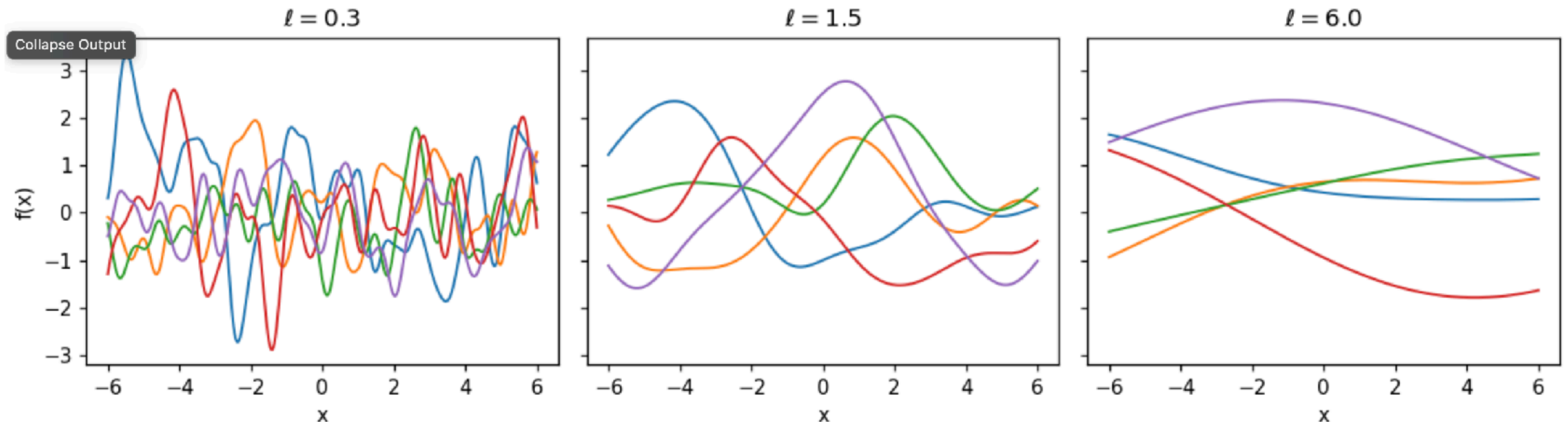


$$k(x, x') \rightarrow \frac{\sigma_w^2 \ell \sqrt{\pi}}{\Delta c} \exp \left[-(x - x')^2 / 4\ell^2 \right] \text{ (square-exponential/radial basis function kernel)}$$

a gaussian process — $f \sim GP(0, k(x, x'))$
 $\{f(x_1) \dots f(x_n)\} \sim \mathcal{N}(0, K)$

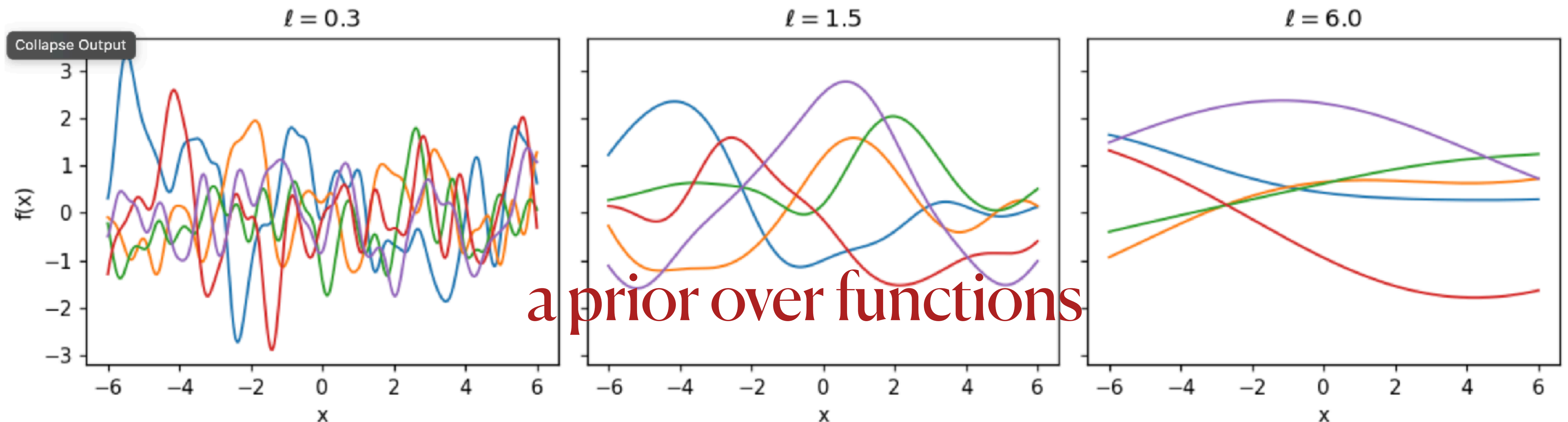
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why is this useful? — gaussian processes for **regression**

GP $\rightarrow$ prior over functions

Bayes $\rightarrow$ posterior over functions given data

gaussian processes for regression — you observe **noiseless**
data $\mathbf{f} = \{f(x_1) \dots f(x_n)\}$

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K — how the observations covary with each other

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$$\{f(x_1) \dots f(x_n), f(x_*)\} \sim \mathcal{N}\left(0, \begin{bmatrix} K & K_* \\ K_*^T & K_{**} \end{bmatrix}\right)$$

K_* — $k(x_i, x_*)$ how strongly each observation is tied to k_*

K_{**} — $k(x_*, x_*)$ prior variance at x_*

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$$p \left(f(x_*) \mid x_*, \mathbf{f} \right)$$

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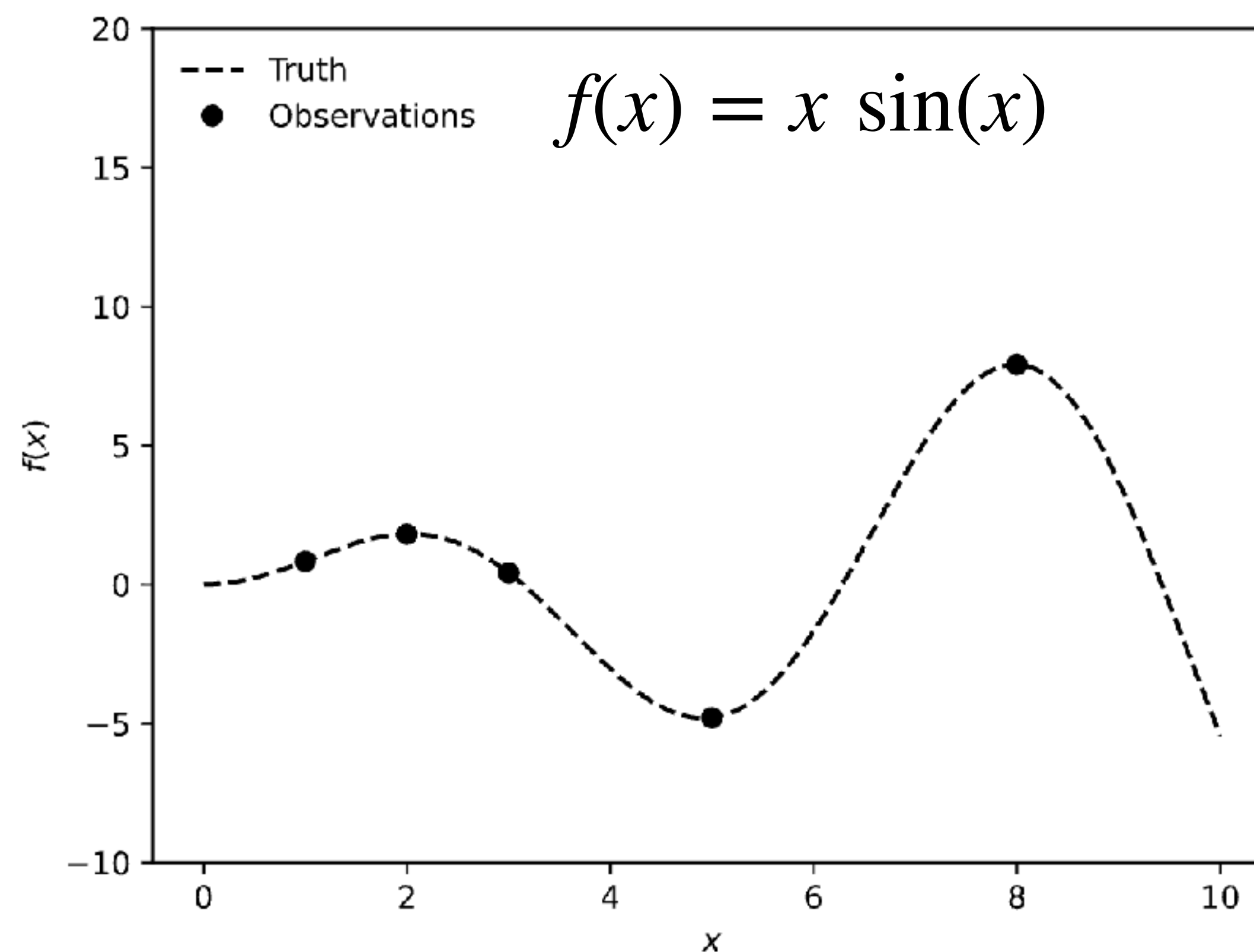
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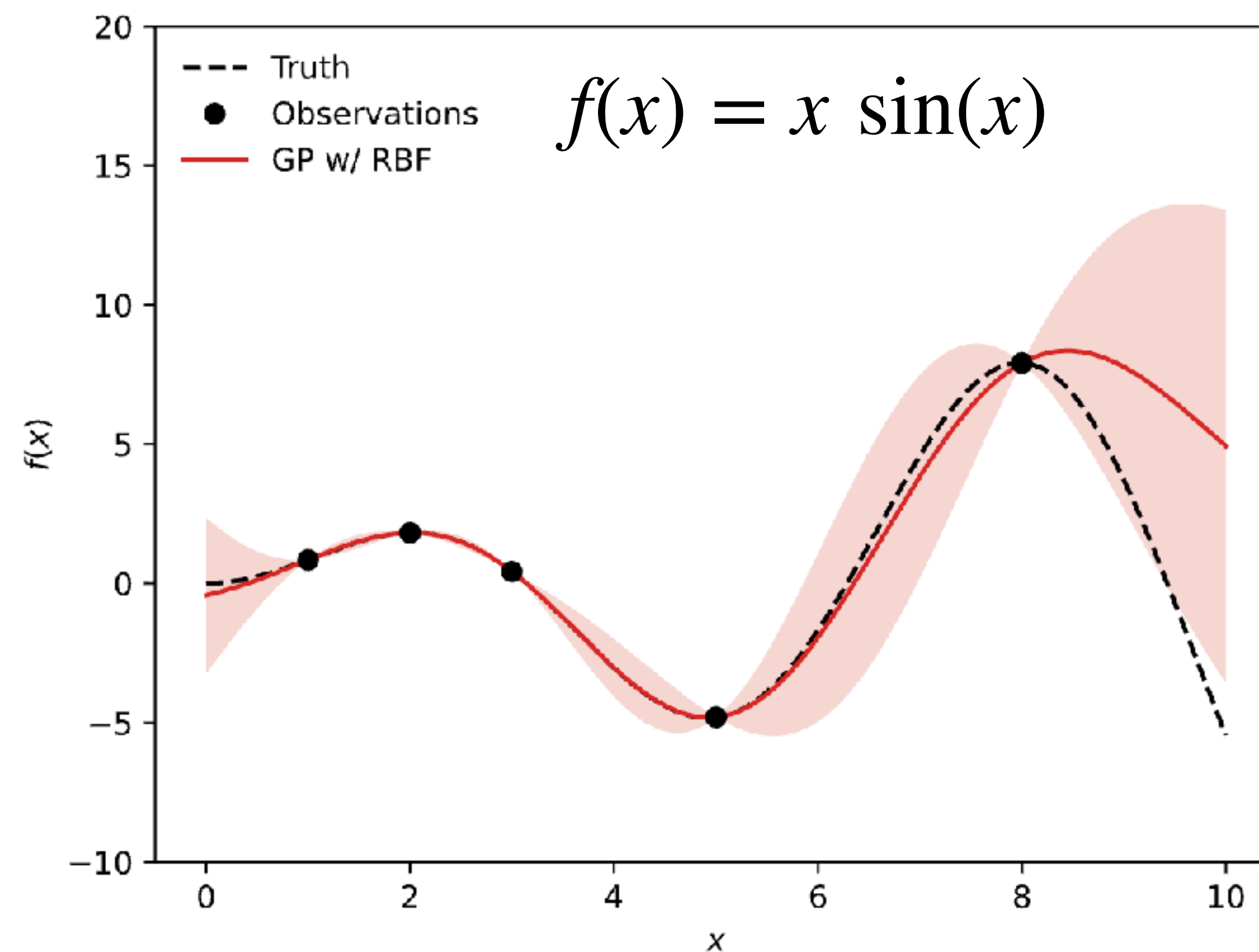
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by Gaussian identity — $x = \begin{pmatrix} x_a \\ x_b \end{pmatrix}$ $\mu = \begin{pmatrix} \mu_a \\ \mu_b \end{pmatrix}$ $\Sigma = \begin{pmatrix} \Sigma_a \Sigma_c \\ \Sigma_c^\top \Sigma_b \end{pmatrix}$ then $p(x_a \mid x_b) = \mathcal{N}(\mu_{a|b}, \Sigma_{a|b})$ where $\mu_{a|b} = \mu_a + \Sigma_c \Sigma_b^{-1} (x_b - \mu_b)$
 $\Sigma_{a|b} = \Sigma_a - \Sigma_c \Sigma_b^{-1} \Sigma_c^\top$

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gaussian processes for regression — you observe **noisy** data

$$y_i = f(x_i) + \varepsilon \text{ where } \varepsilon_i \sim \mathcal{N}(0, \sigma_n^2)$$

$$\{\mathbf{y}, f(x_*)\} \sim \mathcal{N} \left(0, \begin{bmatrix} K + \sigma_n^2 \mathbf{I} & K_* \\ K_*^T & K_{**} \end{bmatrix} \right)$$

$$p \left(f(x_*) \mid x_*, \mathbf{y} \right) = \mathcal{N} \left(K_*^T (K + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y}, K_{**} - K_*^T (K + \sigma_n^2 \mathbf{I})^{-1} K_* \right)$$

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GP predicted variance doesn't depend on $\mathbf{y}$

gaussian processes

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mean prediction: $\overline{f(x_*)} = K_*^T (K + \sigma_n \mathbf{I})^{-1} \mathbf{y}$

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$$a = (K + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y} \quad \overline{f(x_*)} = \sum a_i k(x_i, x_*)$$

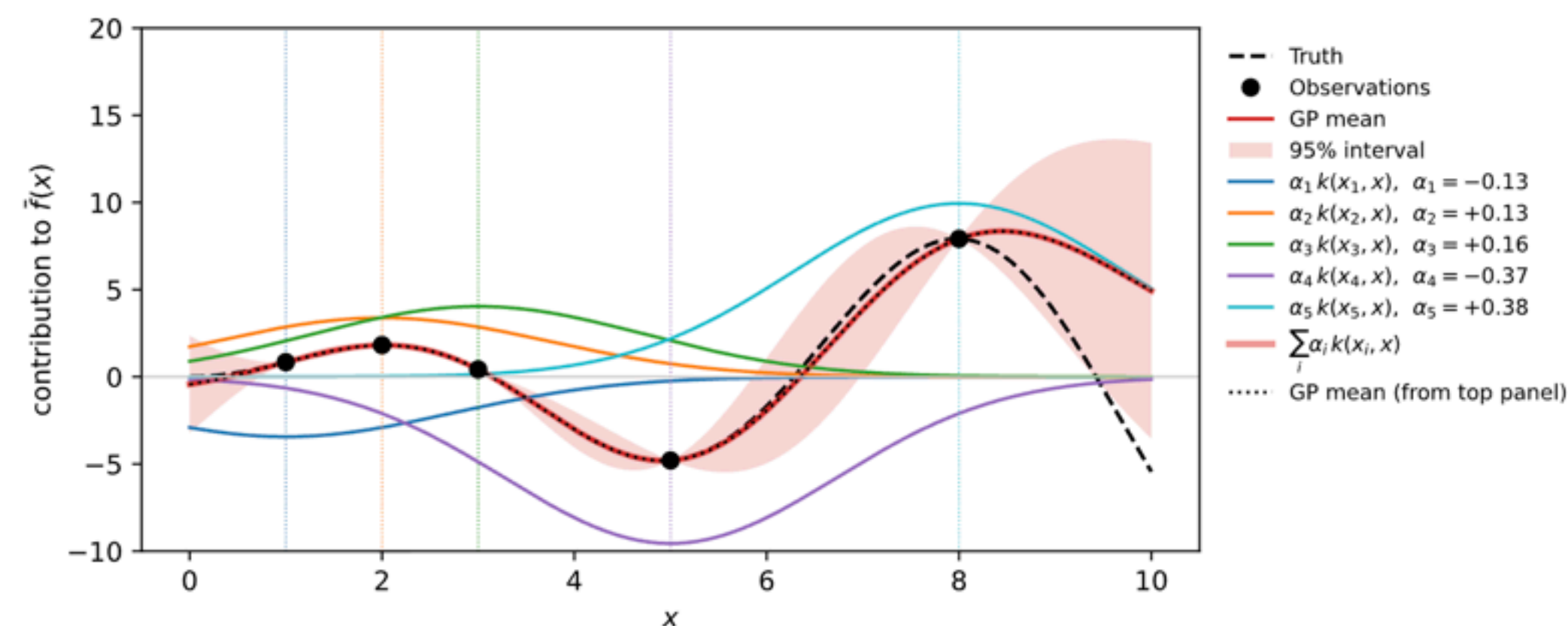
one kernel bump per observation, centered on that observation, weighted by what the data chose

gaussian processes

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for gaussian processes, **kernel is the model**

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kernel is the inductive bias that says which functions are plausible *before you see data*

kernel is set by inductive bias but **what about ℓ and σ_f ?**

$$(\ell, \sigma_f) \rightarrow f \rightarrow y$$

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the GP prior $f \sim \mathcal{N}(0, K)$ where K is
built from ℓ and σ_f and observed data

kernel is set by inductive bias but **what about ℓ and σ_f ?**

$$(\ell, \sigma_f) \rightarrow \boxed{f \rightarrow y}$$

noise model

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we can integrate it out

$$p(\mathbf{y} \mid \ell, \sigma_f) = \int p(\mathbf{y} \mid \mathbf{f}) p(\mathbf{f} \mid \ell, \sigma_f) d\mathbf{f}$$

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$$\log p(y \mid \ell, \sigma_f) = -\frac{1}{2} \mathbf{y}^T K^{-1} \mathbf{y} - \frac{1}{2} \log |K| - \frac{n}{2} \log 2\pi$$

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in-class: population inference using hierarchical Bayes

ast384c/inclass/lec11_gp/gp_lightcurve.ipynb